

Problem #3: Translating Welsh (15 marks)

Machine Translation is a very important technology on which computer scientists and linguists have been working together for many years now. The aim of this technology is to develop computer systems that can automatically translate sentences from one language to another. Though machine translation is a very difficult problem, there are systems that can solve this problem reasonably well. Given below are 8 input English sentences and their corresponding output Welsh translations generated by one such well-known machine translation system developed by *Google*.

1. The man kills the ants.
Mae'r dyn yn lladd y morgrug.
2. The ant kills.
Mae'r morgrugyn yn lladd.
3. The men have killed.
Mae'r dynion wedi lladd.
4. The dog jumped on the man.
Mae'r ci yn neidio ar y dyn.
5. The ants will be killed.
Bydd y morgrug yn cael eu lladd.
6. The dog was going to kill.
Roedd y ci yn mynd i ladd.
7. The man is going to jump on the ant.
Mae'r dyn yn mynd i neidio ar y morgrugyn.
8. The dog will be sent.
Bydd y ci yn cael ei anfon.

Welsh is the native and official language of Wales. It has almost 700,000 speakers, mostly from England and Wales.

a, e, i, o, u, w and **y** are vowels in Welsh. **y** is pronounced as “i” in “bit”; **w** is pronounced as “oo” in “book”; **c** is pronounced as in “case”, **dd** is pronounced as “th” in “this”, whereas **d** is pronounced as in “day”; similarly, **l** and **ll** are pronounced differently. An apostrophe (as in *Mae'r*) indicates a stress on a syllable, and a hat (e.g., *ŵ*) indicates a long vowel.

Assignment 1:

What do you think would be the output of this translation system for the following English sentences?

- a) The ants have killed the man.
- b) The dog jumps on the men.
- c) The men were going to jump on the ants.

Assignment 2:

Can you guess the English input sentences to the machine translation system that led to following Welsh translations? If you think that it is not possible to guess the exact English input for some of the words or sentences, then explain why you think so, and what the possible inputs could be.

- d) *Mae'r ci yn lladd.*
- e) *Mae'r dyn yn anfon SMS.*
- f) *Mae'r cŵn yn mynd i gael ei anfon.*